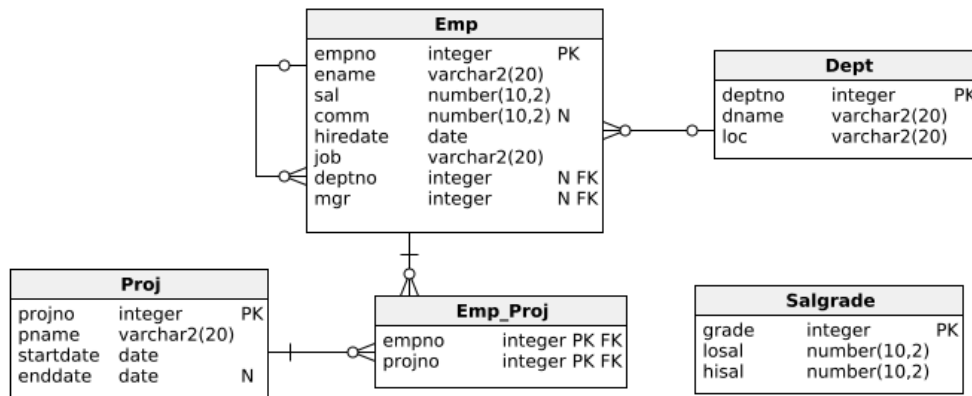


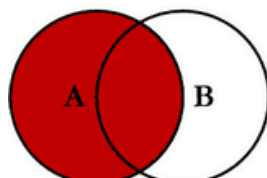
SQL - WHERE, JOIN

Content

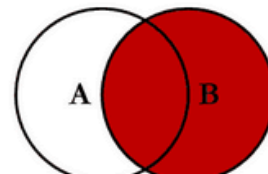
- Connection to Oracle server
- Basic SELECT instructions:
 - SELECT, FROM, ORDER BY
 - WHERE, LIKE, LENGTH, EXTRACT, NULL, NVL, AS
- Joining tables
 - different types of JOIN
 - WHERE



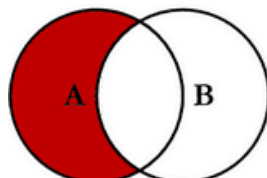
SQL JOINS



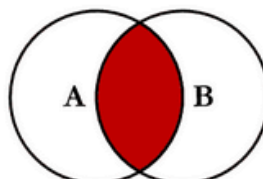
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key



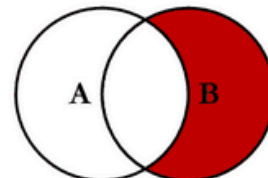
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key



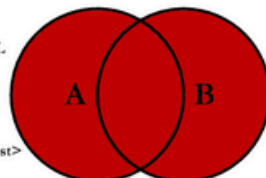
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
WHERE B.Key IS NULL



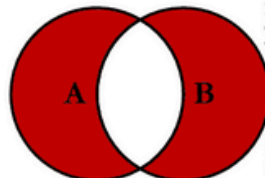
SELECT <select_list>
FROM TableA A
INNER JOIN TableB B
ON A.Key = B.Key



SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL



SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key



SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
OR B.Key IS NULL

Example tasks

1|

List the names, employee numbers, job titles, salaries, and department numbers of all employees with a CLERK job

2|

Find the names and numbers of employees whose bosses are 7902, 7566, or 7788.

3|

Find the four-letter names of the employees.

4|

Find the data of those employees who have a boss.

5|

List the names, annual salary, and commission of all those SALESMANs whose monthly salary exceeds commission. Sort the results by decreasing salary, then by name (ascending).

6|

Select the names of all employees, and the names of the departments in which they are employed. Sort the returned records in alphabetical order of the surnames.

7|

List information about employees whose salaries correspond to salary class 3.

8|

List employees who have worked at the company for more than 5 years.

9|

For each project, list the name and work of all employees in one column.

Individual tasks

1|

List the names of employees whose commission exceeds their monthly salary.

2|

Find the data of those employees whose salaries are outside the range <1000, 2000>.

3|

Find data of those employees who are employed in the position of CLERK, or their earnings are within the range <1000,2000>.

4|

Find all those employees who are employed in the position of MANAGER, or in the position of SALESMAN with a salary above 1500.

5|

List all the data from the SALGRADE table.

6|

Find the data of those employees employed in the position of CLERK whose salaries are not in the range <1000,2000>.

7|

List all distinct (non-repeating) job titles.

8|

List the names, positions, and department numbers of all employees in Department 20 employed at the CLERK position.

9|

List the name, position, and salary of employees who have a boss.

10|

For employees with a monthly salary over 1500, list their names, the locations of their departments, and the names of those departments.

11|

Select employees in department 30 and 40 (name, department no., department name, location). Also list the information of the department, 40 in which there are no employees.

12|

Output the names of all employees and the names of all departments. The result is supposed to show the names of employees not assigned to any departments, as well as the names of departments with no employees.

Additional tasks

1|

List the employees and the names of their departments for people whose salary is outside the range (2000,3000) and whose name does not begin with K.

2|

In one column, write the name of the project and the name of the employee. Include employees who are not working on any project.

3|

For each project, count how many days it took. Call the column with the days LENGTH. Show data only for projects started after 2020.

4|

For each employee, specify their income group and give the name of the boss. Exclude employees who do not have a boss.

5|

List all project information. Give appropriate headings: NAME, START DATE, END DATE. Sort the projects by start date ascending and by name descending.